Ann Allergy Asthma Immunol 000 (2025) 1–11

Contents lists available at ScienceDirect

Long-term prophylactic treatment preferences and willingness to switch therapy in individuals with hereditary angioedema

Courtney Olson, MD[*,1]; Michael Lionetti, BA[†]; Christine Poulos, PhD[‡]; Tamara Ray, MBA[§]; Lorena Lopez-Gonzalez, PhD[§]; Sandra Nestler-Parr, PhD[§]; Patrick Gillard, PharmD[§]; Daniel Soteres, MD[‖]

[*] *Beth Israel Deaconess Medical Center, Boston, Massachusetts*
[†] *KJT Group, Inc, Rochester, New York*
[‡] *RTI Health Solutions, Durham, North Carolina*
[§] *BioCryst Pharmaceuticals, Inc, Durham, North Carolina*
[‖] *Asthma & Allergy Associates, PC, Colorado Springs, Colorado*

ARTICLE INFO

Article history:
Received for publication July 24, 2025.
Received in revised form October 28, 2025.
Accepted for publication December 15, 2025.

ABSTRACT

Background: Long-term prophylaxis (LTP) can help manage hereditary angioedema (HAE). With increasing LTP treatment options, understanding patients' preferences is important for shared decision-making.
Objective: To understand the way individuals with HAE assess the importance of LTP treatment attributes, their LTP treatment preferences, and the impact of disease and treatment attributes on their willingness to switch LTP.
Methods: We conducted an online survey in 2023 among US adults (aged ≥18 years) self-reporting an HAE diagnosis and receiving treatment (LTP, on-demand, or both) or experiencing at least 1 attack in the past 3 months. Best-worst scaling and a discrete choice experiment assessed LTP preferences. A behavior change model assessed willingness to switch LTP.
Results: A total of 150 individuals completed the survey. Respondents rated effectiveness in preventing attacks and reducing the severity of attacks as the most important LTP attributes. Route of administration and convenience were more than twice as important as dosing frequency. Individuals preferred oral daily therapy to biweekly (54% vs 46%) or monthly injections (54% vs 46%). Most individuals (71%) were at least somewhat willing to switch LTP treatments in the next 6 months, particularly those whose HAE was not well controlled, were anxious about taking LTP, were burdened by treating their HAE, or preferred oral administration.
Conclusion: Effectiveness was the primary driver of LTP preference; other factors were also important, including convenience. When effectiveness was equivalent, oral administration was preferred to injectable administration. Individuals with HAE were moderately willing to switch their LTP. By better understanding patients' treatment preferences, health care professionals can individualize LTP recommendations.
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Introduction

Hereditary angioedema (HAE) is a rare genetic disorder characterized by recurring episodes of swelling in various body parts, such as the limbs, face, abdomen, gastrointestinal tract, and larynx, that can last several days and can be life-threatening.[1-3] There are 3 types of HAE: HAE with C1 esterase inhibitor (C1INH) deficiency (type 1 or 2) and HAE without C1INH deficiency.[3-5] Type 1 and type 2 HAE are primarily caused by the lack of or reduced function of C1INH, respectively.[2,3]

The substantial negative impact of HAE on quality of life can affect all aspects of daily living as a result of unpredictable attacks and the chronic nature of the disease.[4,6-9] Individuals with HAE also have a higher rate of depression and anxiety.[4,9-11] Moreover, HAE is associated with increased absenteeism, lost productivity, and higher health care costs, primarily due to emergency department visits and hospital stays.[4,7,9,12,13]

Treatment for HAE includes on-demand therapy for acute attacks, short-term prophylaxis, and long-term prophylaxis (LTP) for chronic disease management.[14-16] LTP reduces the frequency and severity of HAE attacks and helps individuals with HAE achieve disease control and normalize their lives.[17-19] Treatment for HAE can be burdensome owing to administration challenges,

Disclaimer: This work was previously presented in part at the American College of Allergy, Asthma, and Immunology, October 24 to 28, 2024, in Boston, Massachusetts, and at the American Academy of Allergy Asthma & Immunology, February 28 to March 3, 2025, in San Diego, California.

[2] Patient author.
Address correspondence to: Courtney Olson, MD, Beth Israel Deaconess Medical Center, 330 Brookline Ave, Boston, MA 02215 E-mail: courtney.n.olson@cuanschutz.edu.

https://doi.org/10.1016/j.anai.2025.12.011
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accessibility issues, or adverse effects that have physical, emotional, social, or financial consequences.[20-22] Current LTP options include intravenous, subcutaneous, or oral treatment, with varying frequency of administration.[23]

Disease burden and perceived treatment effectiveness are important factors in LTP decision-making and may affect an individual's willingness to start or switch prophylactic therapy.[24-26] Individuals evaluate satisfaction with their prophylactic therapy on the basis of factors such as attack reduction, symptom relief, and treatment convenience.[27-29] Expert consensus and the World Allergy Organization/European Academy of Allergy and Clinical Immunology guidelines for HAE recommend shared decision-making and individualized management plans and stress the importance of considering patients' health-related quality of life and personal preferences in LTP treatment decisions.[19,30,31] It has been recommended that patient empowerment be a focus of health care professional (HCP) decision-making in partnership with their patients.[32-34]

According to the World Allergy Organization/European Academy of Allergy and Clinical Immunology guidelines, HCPs should discuss LTP options with their patients at every visit.[19] Regular discussions about starting or switching LTP are important because an individual's satisfaction with their treatment can affect adherence; treatment needs or preferences may change over time as disease severity changes and new treatment options become available.[18,35,36] Optimal conversations include ways each therapy can address clinical factors, disease burden, and treatment burden, ensuring individuals' preferences are considered.[19,24-26,30] As more LTP treatment options are available, understanding patients' prophylactic treatment preferences and willingness to switch therapies is important for shared decision-making. However, the literature assessing the LTP treatment preferences of individuals with HAE is limited.[37,38]

The objectives of this study were to (1) quantify the relative importance of HAE LTP attributes in treatment choice and ways individuals with HAE prioritize LTP treatment features or benefits; (2) determine individuals' preference for oral or injectable LTP; and (3) assess the influence of disease burden and perceived treatment effectiveness on individuals' motivation to switch therapies and willingness to switch LTP.

Methods

Study Design

We conducted a cross-sectional study with 2 phases: qualitative interviews and an online survey preceded by pilot testing. The study followed methodologic best practices for conducting research with patients and assessing patient preferences, including attribute selection and pretesting.[39-43] The qualitative research was conducted from May 26 to June 20, 2023, to inform the survey design. Data from the online survey were collected from October 12 to 27, 2023 in the United States by a third-party vendor (KJT Group, Inc, Rochester, New York) using 2 online research panels (KJT Group, Inc's custom panel of individuals with HAE and the HAE patient panel of Connected Research & Consulting, LLC, Boca Raton, Florida). Individuals were recruited to be members of the panels through a combination of outreach channels and were screened using a series of confirmatory and vetting procedures; during the registration and enrollment process, they provided detailed demographic and health information, including self-reported diagnosis of HAE and relevant treatment experience. Panel members had agreed to be contacted for research purposes. Interested respondents meeting predefined inclusion criteria (ie, age ≥ 18 years, US residency, and self-reported diagnosis of HAE) were directed to a secure online survey platform (Decipher, Forsta, Stamford, Connecticut). Participants electronically consented to participate before starting the survey. Respondents self-administered the screening questions and were invited to complete the survey if they met all inclusion criteria. The survey was available in English. Participants who completed the survey were offered a modest monetary incentive. This study was deemed exempt from review by the WCG Institutional Review Board.

Participants

Participants were required to reside in the United States, be aged 18 years or older, and self-report a diagnosis of HAE (HAE-C1INH type 1 or 2 or HAE without C1INH deficiency) by an HCP. At the time of the survey, participants had to be using prophylactic and/or on-demand HAE treatments; those not currently treating their HAE but who had experienced at least 1 attack in the past 3 months were also included. A series of screening questions determined respondents' qualification for the survey. After answering a few demographic questions, respondents were presented with a list of several medical conditions and asked whether they had been diagnosed with any of those by an HCP. Respondents who indicated that they had been diagnosed with HAE were asked what type of HAE they were diagnosed with. Those selecting "not sure" were presented with definitions of the 3 HAE types; respondents who confirmed their HAE type continued to the next question in the screener. Respondents were then asked whether they were currently taking any prescription medication(s) for their HAE attacks and whether they were taking medications to prevent HAE attacks. Respondents were excluded if they were unwilling to participate. The final sample size was selected to balance recruitment feasibility and statistical power, particularly for the best-worst scaling and discrete choice experiment (DCE) analyses.[44,45]

Survey Design

The study comprised a 30-minute, self-administered online survey. The survey instrument was designed on the basis of findings from the qualitative interviews with 15 individuals with HAE. The qualitative interviews (1) explored the most important features of LTP to determine ways individuals with HAE rank importance of different features and benefits; (2) assessed ways perceived treatment and disease burden affect patients' willingness to switch prophylactic therapy and ways product features are prioritized; and (3) determined individuals' motivations to consider a prophylactic therapy switch (or start) and voice their preferences to HCPs.

The survey comprised binary (yes and no), numeric entry, single-select, multiple-choice, open-text, and Likert scale questions about HAE diagnosis, disease and treatment history, disease burden, well-being, quality of life, disease severity, experience with LTP, attitudes and beliefs about treatment and treatment burden, barriers to and drivers of LTP use, behavior change stage, and willingness to switch medications. Likert scales (7-point and 11-point) included agreement, severity, and satisfaction assessments. Response options of "other" or "not sure" were provided as relevant. Range limits were used to ensure accuracy for numeric entry questions. Well-being was assessed by the World Health Organization Well-Being Index, a brief, widely used, validated scale to assess mental well-being.[46] The Angioedema Quality of Life Questionnaire (AE-QoL) assessed quality of life; this validated instrument was chosen for this study because it is commonly used in clinical trials of HAE treatments.[47]

Two methods were used to assess LTP treatment preferences, following best practices for assessing patient preferences.[39,48-50] Respondents were presented with a best-worst scaling exercise listing 16 prophylactic treatment attributes (shown in 12 sets of 4 items each) one may consider when deciding whether to use a given prophylactic treatment. Individuals were asked to choose 1 item from each set that was most important and 1 least important in treatment choice. The items included in the best-worst scaling exercise were

primarily selected on the basis of findings from the qualitative interviews. A DCE, a research technique to determine preferences based on hypothetical choice sets, was also used.[51] The DCE was a series of choice tasks in which respondents were asked to choose between 2 hypothetical, experimentally designed LTP treatment options (with profiles comparing 1 oral and 1 subcutaneous injection) described by 3 attributes (dosing frequency, average number of attacks over a 3-month period, and change in severity of attacks from what they currently experienced), each with multiple levels. The attributes and levels selected for the DCE were based on information from the qualitative research and available LTP treatment options currently available or in development. The combination of attributes and levels (ie, profiles) that respondents reviewed was determined a priori according to an experimental design (using CBC Hierarchical Bayes, v5.5.6, Sawtooth Software, Inc, Provo, Utah). The experimental design was generated using a statistical algorithm for D-efficient fractional factorial designs conforming to the principles of orthogonality, minimal overlap, and level balance. This algorithm takes a predefined number of screens to display and the number of profiles per screen and determines the optimal experimental design. Because the possible number of combinations a patient could view was impossible to capture in a single experimental design, 20 versions of treatment combinations were created. Respondents were randomly assigned to a version, shown 10 screens with 2 possible treatment options per screen, and asked to select their preference. Various dosing frequencies were tested in the DCE; we have reported the frequencies for profiles similar to currently available LTP treatments (approved by the US Food and Drug Administration [FDA] as of October 2023) or those in development and expected to be approved by the FDA in the next 12 months.

Willingness to switch LTP was directly assessed by an 11-point Likert scale of "not at all willing" to "extremely willing." Individuals currently taking an LTP were asked about their willingness to switch their LTP. Individuals not currently taking an LTP were asked about their willingness to start an LTP. "Switch" is used throughout for brevity. A modified version of the transtheoretical model of health behavior change assessed willingness to switch.[52] This widely accepted model suggests that health behavior change involves progression through 6 stages of change: precontemplation, contemplation, preparation, action, maintenance, and termination.[52] Individuals' perception of control over their treatment options was assessed using a series of questions to assess behavioral intention following the theory of planned behavior, which posits that behavior is primarily driven by an individual's intentions and influenced by 3 factors: attitudes toward the behavior, subjective (social) norms, and perceived behavioral control.[53,54]

Before data collection, the survey instrument was pilot-tested with 9 people with HAE meeting the study inclusion criteria. Participants in the pilot test reviewed the survey using a think-aloud pretest through web-assisted telephone interviews. These interviews were conducted to assess the face validity and patient centricity of the survey instrument (eg, ensuring the questions were relevant, appropriately phrased, and understandable). Minor revisions were made to the survey instrument on the basis of feedback from the pilot testing.

Statistical Analysis

Descriptive statistics were performed, including mean (SD) and median (IQR) for continuous variables and proportion for categorical variables. Statistical analyses of the aggregated data were conducted using R Statistical Software (R Core Team 2021, Vienna, Austria), CBC Hierarchical Bayes, Q Research Software for Windows (a division of Displayr, Inc, Pirmont, Australia), Stata (version IC 14.2, StataCorp LLC, College Station, Texas), and Excel (Microsoft, Redmond, Washington). Statistical significance was set at P less than .05, using 2-tailed tests. Survey responses for each study participant who completed the survey and met predefined quality control measures were included in the analyses; thus, there were no missing data. Outliers were identified and truncated to the median as appropriate, determined on a case-by-case basis.

Data from the best-worst scaling exercise were analyzed using a hierarchical Bayesian multinomial regression model to generate each attribute's mean relative importance score. Relative importance scores ranged from 0 to 100 (summing to 100) and were ratio-scaled (ie, an attribute with a rating of 20 was twice as important as one with a rating of 10). Data for the DCE were analyzed using a hierarchical Bayesian multinomial regression model that used Markov Chain Monte Carlo methods for estimating parameters. Posterior estimates are taken from the distribution of Markov Chain Monte Carlo draws when the model shows appropriate convergence. The regression model results were used to simulate preference shares for treatment profiles that vary by mode of administration and frequency. This model informed trade-off preferences for HAE LTP treatments.

Behavioral intention analysis was used to evaluate the relative influence of the 3 domains of the theory of planned behavior on an individual's decision to switch prophylactic therapy. Individuals' level of agreement with various statements was mapped back to behavioral attitudes (ie, beliefs about medication effectiveness and convenience, confidence in taking, quality of life with their current or a different medication, and the risk/hassle of switching medications), social norms (ie, HAE medication use/experiences of other family members or patients with HAE), and perceived control (ie, concerns about HAE control with another medication, and ease/difficulty of switching medication because of insurance), which served as the inputs or possible drivers of overall willingness to switch their current therapy; these were transformed into composite scores and then assessed through a regression model. Internal reliability for each composite was tested through Cronbach's α. Items lacking internal consistency with the composite were removed. Drivers were run in total with inputs coming from or within each domain; these domain-specific, ordinary least squares linear regression models were conducted to determine the extent to which the tested behavioral constructs may affect the decision to engage in prophylactic therapy switching while controlling for the impact of the other 2 domains.

Results

Baseline Characteristics of the Study Population

Invitations were sent to 390 potential respondents from the online panels, and 192 individuals responded, for a response rate of 49%. A total of 150 individuals with HAE qualified for and completed the survey, most of whom had type 1 HAE (Table 1). The mean age was 47 years; most respondents were female (81%). Most patients (83%) had at least 1 of the listed comorbid conditions. Nearly all respondents (n = 148, 99%) were taking prescribed HAE medications, primarily prophylactic medication, with or without on-demand treatment. Among those taking LTP (n = 139), 92% were on an injectable medication; 7% were on oral therapy, and 1% were taking both injectable and oral LTP. Respondents not taking LTP were using on-demand treatment only (n = 9) or were not taking any HAE treatment (n = 2).

Relative Importance of Hereditary Angioedema Long-Term Prophylaxis Treatment Attributes

According to the best-worst scaling exercise, LTP treatment attributes rated as most important in treatment choice were effectiveness in preventing attacks (16.9) and effectiveness in reducing the severity of attacks (16.1; Fig 1). Administration location (eg, home, hospital, or

Table 1
Demographic and Clinical Characteristics of Individuals With HAE

Characteristics	Individuals with HAE (N = 150)
Region, n (%)	
South	63 (42.0)
Midwest	33 (22.0)
West	28 (18.7)
Northeast	26 (17.3)
Age, y, mean (SD)	47.1 (13.4)
Time since diagnosis, y, mean (SD)	26.8 (12.8)
HAE type, n (%)	
Type 1	116 (77.3)
Type 2	27 (18.0)
HAE with normal C1-Inhibitor	7 (4.7)
HAE attacks	
Mean (SD)	
Past 1 mo	1.1 (1.5)
Past 3 mo	3.4 (4.5)
Past 6 mo	6.8 (10.3)
Median (IQR)	
Past 1 mo	1 (0-2)
Past 3 mo	1 (1-4)
Past 6 mo	3 (1-8)
Current HAE treatment, n (%)[a]	
Prophylactic medication with or without on-demand medication	139 (92.7)
Injectable LTP (subcutaneous or intravenous)	128 (92.1)
Oral LTP	10 (7.2)
Both injectable and oral LTP	1 (0.7)
On-demand medication only	9 (6.0)
Not taking any HAE treatment but experienced ≥1 attack in the past 3 mo	2 (1.3)
Gender, n (%)[a]	
Female	121 (80.7)
Male	28 (18.7)
Transgender	0 (0.0)
None of these	1 (0.7)
2022 household income, US dollars, n (%)	
<39,999	35 (23.3)
40,000-74,999	38 (25.3)
75,000-149,999	40 (26.7)
150,000-249,999	10 (6.7)
≥250,000	13 (8.7)
Declined to answer	14 (9.3)
Medical specialty of physician who treats HAE, n (%)[a]	
Allergy/immunology	139 (92.7)
Primary care/general/family practice	7 (4.7)
Internal medicine	3 (2.0)
Rheumatology	1 (0.7)
Covered by health insurance, n (%)	
Yes	147 (98.0)
No	3 (2.0)
Health insurance type, n (%)[b]	
Commercial/private insurance	108 (72.0)
Medicare	37 (24.7)
Medicaid	15 (10.0)
Other	6 (4.0)
Not sure	2 (1.3)
Uninsured	3 (2.0)
Health insurance plans including prescription drug coverage, n (%)[a]	
Yes	142 (94.7)
No	1 (0.7)
Not sure	4 (2.7)
Uninsured	3 (2.0)
Living family members diagnosed with HAE, n (%)[a]	
0	22 (14.7)
1-5	85 (56.7)
6-10	37 (24.7)
≥11	6 (4.0)
Diagnosed comorbid conditions, n (%)[b]	
Anxiety	58 (38.7)
Depression	47 (31.3)
Obesity	39 (26.0)
High blood pressure	33 (22.0)
High cholesterol/hypercholesterolemia	29 (19.3)
Arthritis	27 (18.0)
Asthma	27 (18.0)
Migraines	21 (14.0)
Anemia	21 (14.0)
Sleep disorders	19 (12.7)
Diabetes	17 (11.3)
Osteoporosis	11 (7.3)
Cancer	10 (6.7)
Kidney disease	7 (4.7)
Heart disease	7 (4.7)
Chronic obstructive pulmonary disease	3 (2.0)
Other	24 (16.0)
None of the above	26 (17.3)

Abbreviations: HAE, hereditary angioedema; LTP, long-term prophylaxis.
[a]Percentages may not sum to 100% owing to rounding.
[b]Percentages may sum to more than 100% owing to multiple responses allowed.

infusion center), adverse effects, out-of-pocket cost, and financial support through patient support programs were similarly important (range: 8.1-9.2), followed by long-term risks (6.8). Route of administration (6.1) and convenience/ease of use (6.3) were more than twice as important as dosing frequency (2.7).

Long-Term Prophylaxis Treatment Preference—Discrete Choice Experiment

When presented with the option of equally effective oral or subcutaneous LTP (defined as an average of 3 attacks over a 3-month period), 54% of respondents preferred once-daily oral administration, and 46% preferred subcutaneous injection once every 2 weeks (Fig 2). Oral treatment preference remained the same when compared with a monthly subcutaneous injection (Fig 2). Somewhat more individuals in the sample preferred oral administration, regardless of the route of administration of their current therapy (Table 2). When asked to directly indicate their agreement with a statement about their preference for an oral or an injectable or infused prophylactic medication, 35% had no administration preference; 11% strongly preferred injection or infusion administration, and 25% strongly preferred oral administration.

Willingness to Switch Long-Term Prophylaxis Treatments

Most individuals with HAE were at least somewhat willing to switch LTP treatments in the next 6 months (rating of 3-7; Fig 3A). Treatment burden and administration preference contributed to individuals' willingness to switch their LTP. Individuals who reported being anxious about taking LTP medication or who found treating HAE to be a burden were more willing to switch treatments. Disease burden also affected individuals' willingness to switch LTP treatments (Fig 3B). Individuals reporting higher burden (AE-QoL > 30) were more willing to switch LTP than were those with low perceived burden (AE-QoL ≤ 30). Individuals reporting greater perceived control over their HAE attacks were less willing to switch their medication.

The relationship between respondents' agreement with 14 statements related to the impact of HAE on overall quality of life was assessed. The only statement significantly correlated with willingness to switch LTP was "I often worry about my HAE" (eTable 1). The behavioral intention analysis showed that behavioral attitudes and social norms influenced individuals' willingness to switch LTP more than perceived control (Fig 4A and B). Significant influencers of willingness to switch LTP included perception of effectiveness, convenience, positive word of mouth, and quality of life (eTable 2). Belief

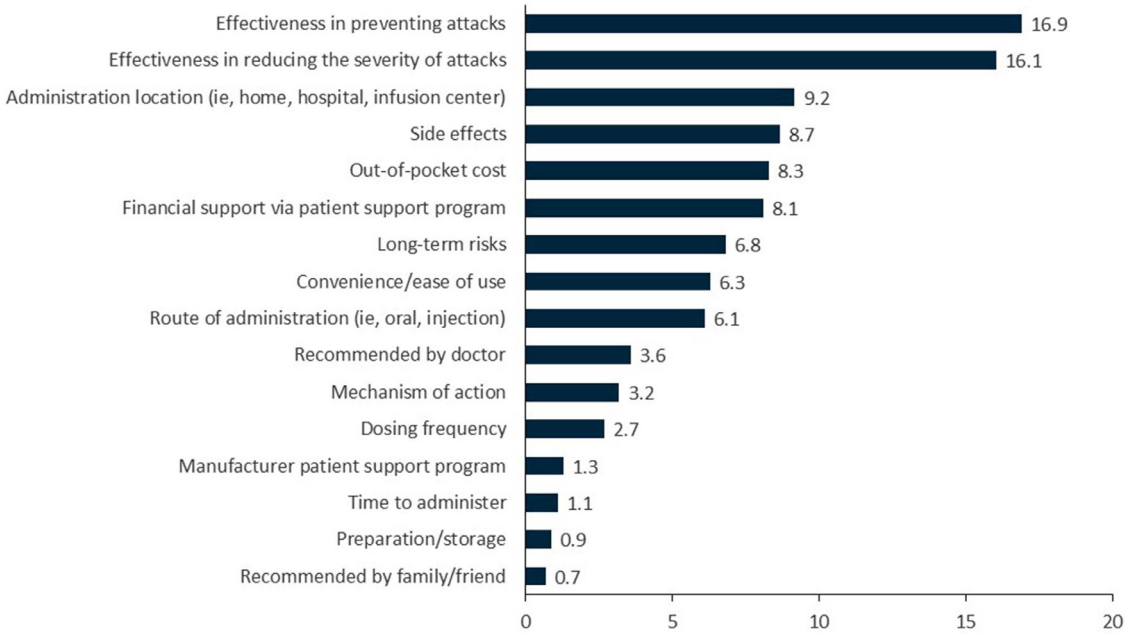


Figure 1. Relative importance of factors in LTP treatment among individuals with HAE (best-worst scaling exercise). Data from the best-worst scaling exercise were analyzed using a hierarchical Bayesian multinomial regression model to generate each attribute's mean relative importance score. Relative importance scores summed to 100 and were ratio-scaled (ie, an attribute with a rating of 20 was twice as important as one with a rating of 10). HAE, hereditary angioedema; LTP, long-term prophylaxis.

that oral medication is as effective as or more convenient than an injectable medication explained 23% and 16% of overall willingness, respectively. Individuals who had observed or were exposed to positive patient testimonials through a support group, social media group, or other means would be more interested in switching (explaining 9% of overall willingness). Conversely, those who believed that their quality of life was acceptable enough to prevent them from taking the risk of switching to a different medication were significantly less willing to switch medications, explaining 30% of their overall willingness to switch their medication.

Discussion

This study indicates that although effectiveness in preventing attacks and effectiveness in reducing the severity of attacks were the top drivers of LTP treatment preference, individuals with HAE also considered other factors important to their definition of treatment preference. These factors included administration location (at home or a health care setting such as a clinic or hospital), adverse effects, cost/financial support, long-term risks, convenience, and route of administration. Among a patient population mostly on injectable LTP (92%), when effectiveness was equivalent, slightly more than half of

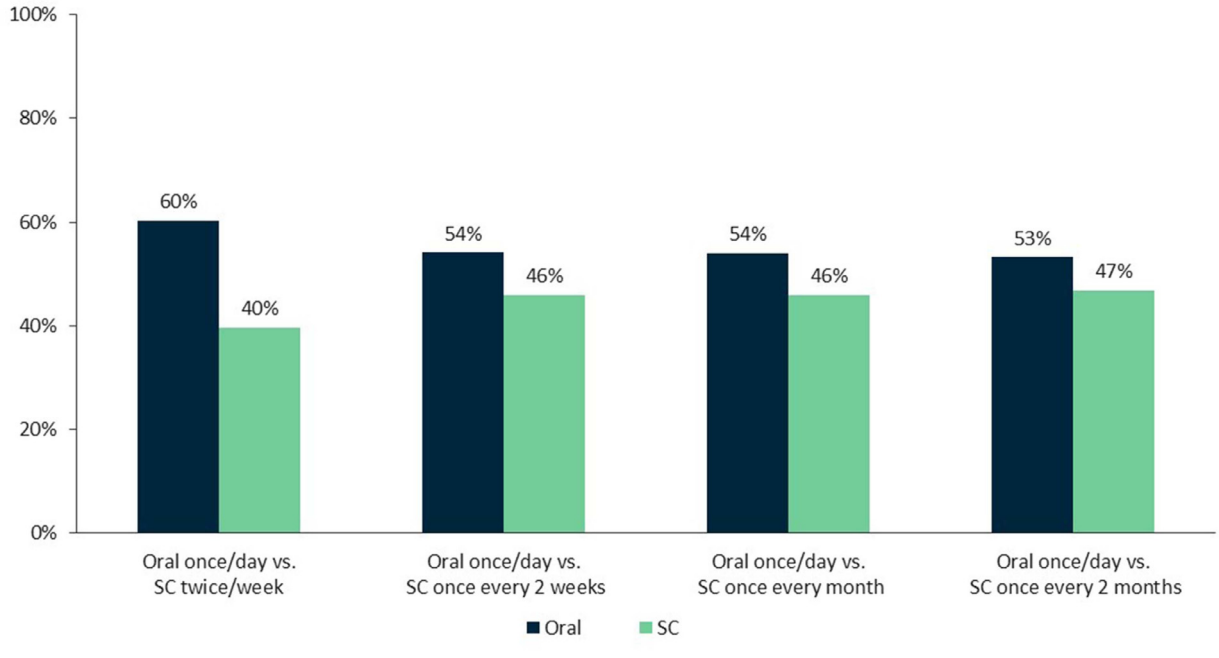


Figure 2. Simulated preference shares based on discrete choice experiment regression analysis by administration type and frequency (N = 150). Preference share is the same for each comparison regardless of attack frequency (ie, 0, 3, or 6 attacks over a 3-month period). HAE, hereditary angioedema; LTP, long-term prophylaxis; SC, subcutaneous.

Table 2
Simulated Preference Shares Based on Discrete Choice Experiment Regression Analysis by Current Long-Term Prophylaxis Treatment (N = 150)

Current treatment	Oral (once/d dosing, 3 HAE attacks over a 3-mo period), %	Injection (once/2-wk dosing, 3 HAE attacks over a period), %
Lanadelumab (n = 76) Subcutaneous injection every 2 wk	53.0	47.0
C1 esterase inhibitor, subcutaneous (n = 32) Subcutaneous injection every 3-4 d	56.9	43.1
Berotralstat (n = 9) Oral administration every day	60.3	39.7
Other (n = 33)[a]	52.2	47.8

Abbreviation: HAE, hereditary angioedema.
[a]Other medications include androgens, C1 esterase inhibitor (human) (intravenous injection), on-demand medications taken prophylactically, clinical trial drugs, or no medication taken prophylactically.

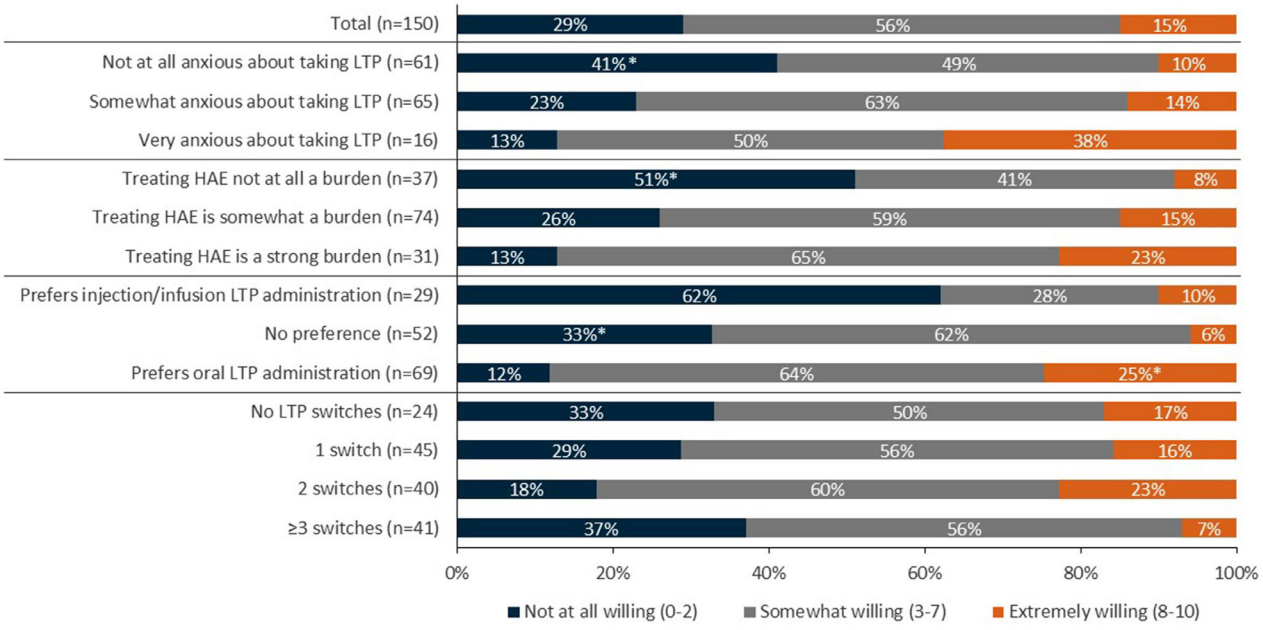


Figure 3A

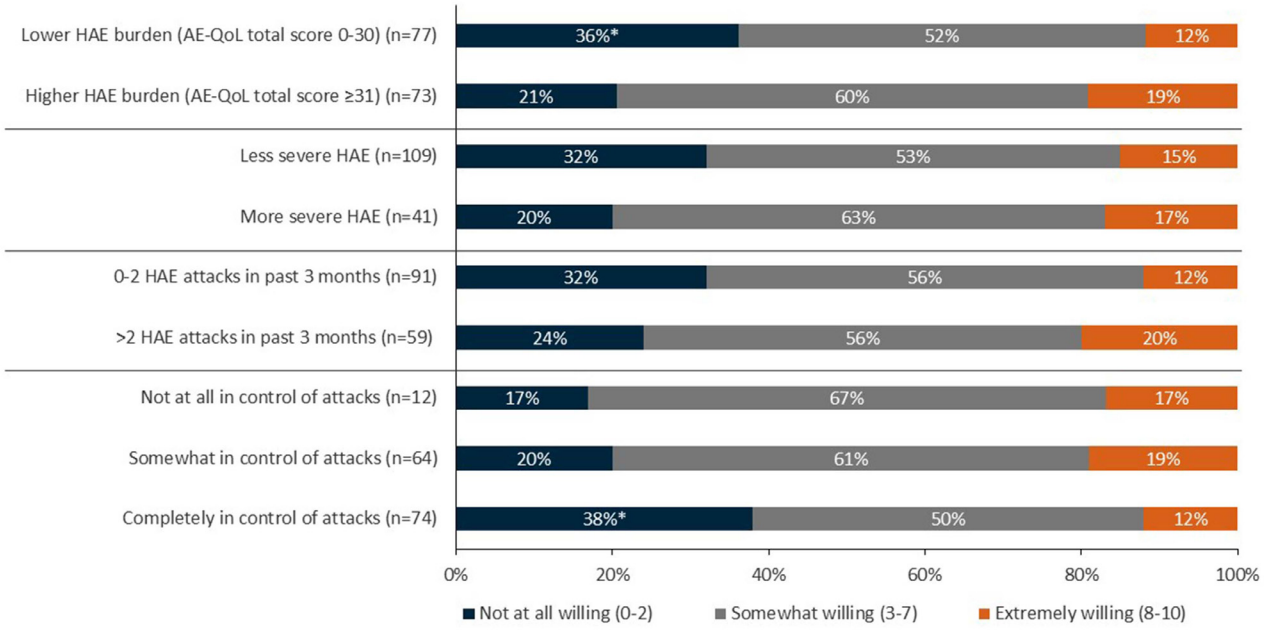


Figure 3B

Figure 3. Willingness to switch HAE LTP medications in the next 6 months overall and by treatment burden and preferences (A) and by disease burden (B). Willingness to switch LTP was rated on an 11-point Likert scale ranging from 0 to 10. *P is less than .05 compared with the comparator group(s) for the same scale category (for the stratifications with n ≥ 30). AE-QOL, Angioedema Quality of Life Questionnaire; HAE, hereditary angioedema; LTP, long-term prophylaxis.

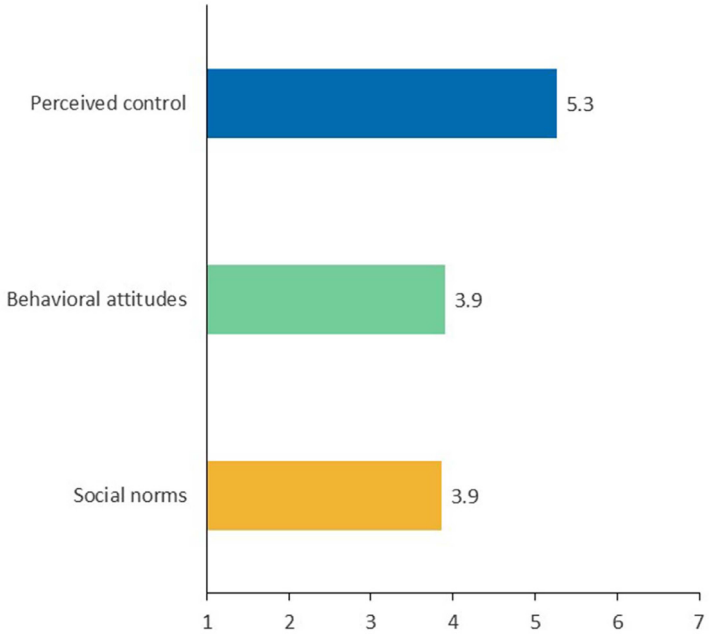


Figure 4A

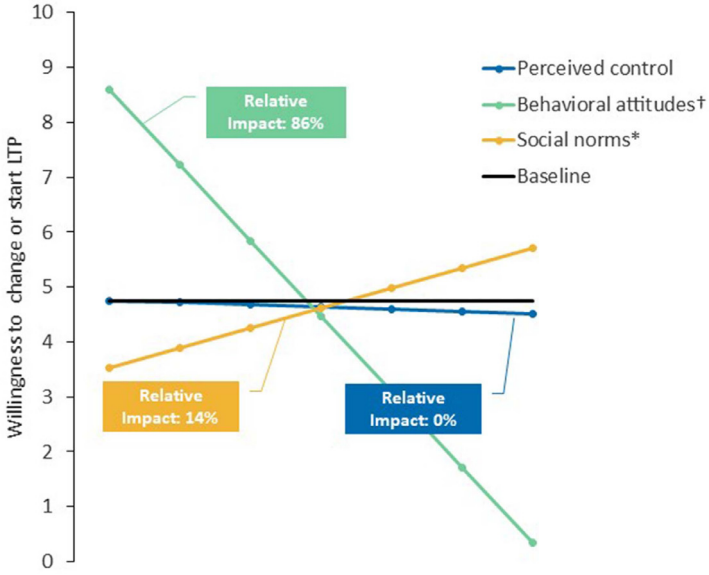


Figure 4B

Figure 4. Behavioral intention analysis domain performance (A) and behavioral intention analysis domain impact on willingness to switch LTP (B) (N = 150). *P is less than .01, †P is less than .001. Composite scores on a scale in which "1" means "strongly disagree" and "7" means "strongly agree." HAE, hereditary angioedema; LTP, long-term prophylaxis.

individuals with HAE preferred a once-daily oral LTP to an injectable administered twice a week, once every 2 weeks, or monthly. Oral administration was preferred to subcutaneous injection regardless of individuals' current treatment. Individuals with HAE were moderately willing to switch LTP, particularly those with greater disease and treatment burden. These findings show that LTP preference is based on many factors and is affected by each individual's experiences with their HAE disease and treatment.

Our research findings highlight the importance of understanding individuals' preferences for LTP treatment in clinical practice and shared decision-making between patients and their HCPs.[55,56] Individuals' preferences in choosing treatments for their conditions are becoming increasingly important for HCPs, patients, and regulatory bodies. Patient-focused drug development efforts by the US FDA emphasize the need to include patient experiences and priorities in developing and evaluating medications, including disease and treatment burden.[57] Patient preference should be a key consideration in treatment selection, along with other patient-specific, clinical, and medication factors.[19,58]

Our study adds to the limited body of literature assessing the LTP preferences of individuals with HAE.[37,59,60] At the 2017 Voice of the Patient Summit, individuals with HAE indicated that the route of administration was the most important driver of treatment preference, and they wanted more convenient treatment options.[60] A series of surveys in 2010, 2013, and 2019 found that few physicians (8%-16%) treating HAE considered convenience an important nonefficacy factor when considering LTP.[14] However, in 2013 and 2019, 27% of physicians recognized this treatment attribute as important to their patients, increasing from 10% in 2010; this likely reflects the availability of new LTP treatment options and a better understanding of the factors affecting patients' quality of life.[14] Moreover, the route of administration increased in importance among the physicians surveyed between 2010 and 2019.[14] In a survey about HAE and its treatments, 86% of individuals with HAE reported satisfaction with their LTP but were still interested in a medication that was easier to administer; 89% learned to tolerate the difficult aspects of their HAE treatment, suggesting that there may be room for improvement in treatment options.[22] These studies underscore the way individuals with HAE value LTP treatment attributes, extending beyond effectiveness.

Because the goal of LTP is to control HAE while minimizing treatment burden and maintaining a normal quality of life,[19] HCPs must consider clinical and treatment burdens when making LTP therapy recommendations.[61] Consideration of treatment burden is critical as HAE treatment continues to expand, providing patients and physicians with more options.[62] Furthermore, treatment preferences change over time, especially in individuals diagnosed with HAE as children. Previous research showed that many patients discuss LTP challenges with their HCPs and find these conversations very helpful.[22] However, a qualitative study exploring ways patients and physicians communicate about HAE found that physicians introduced prophylactic treatment passively, without distinguishing between short-term prophylaxis and LTP.[63] When choosing an LTP therapy, shared decision-making is essential to considering patients' attitudes, preferences, and treatment goals.[32,34,64] Riedl et al[32] found that shared decision-making empowered patients with HAE to be more likely to request a change in their treatment, which can benefit some patients; a small study in 34 individuals with HAE who had switched from injectable LTP to oral LTP reported increased satisfaction and convenience.[28] Therefore, regular conversations between HCPs and their patients with HAE are essential for effective disease management[65] and should address key aspects of disease control and treatment factors, including effectiveness, adverse effects, and route of administration.

Strengths of this study are the use of best practices and methods for assessing patient preferences[39,41] and the use of validated instruments to assess key measures of interest, such as disease burden. Furthermore, the survey was pretested to assess face validity, and inferential and predictive statistical tests were used to analyze survey responses. To reduce potential bias, survey respondents were not informed about the purpose of the study when entering the survey and answering the screening questions. Finally, although geographic diversity was achieved, it is not clear that the sample is representative of the patient population in terms of other characteristics.

This study has limitations, including the use of online panels, sample size, and the self-reported nature of the study. Exclusion lists from the online panels were used to identify and exclude potential duplicates during data collection, and data for any duplicate responders were removed; although these steps greatly minimize the likelihood of overlap, the possibility of overlap between the panels is not eliminated. Recall bias, social desirability bias, and selection bias could limit the generalizability of the data to a broader population of individuals with HAE, as could demographic differences between survey respondents and those who did not respond or do not participate in research panels. The survey was only offered in English, potentially limiting the ability of otherwise qualified individuals to participate in the study. Race and ethnicity were not collected from survey respondents, limiting further insights into the specific characteristics of the study population. Moreover, although HAE similarly affects male and female patients, most individuals in our sample were female. Our survey population primarily comprised individuals taking an injectable LTP (92%), higher than the estimated share (approximately 80%) of injectable medications in the United States. Therefore, our survey may underestimate treatment preferences among individuals taking oral LTP.[66] Furthermore, the survey asked respondents about prophylactic (preventive) medication for HAE but did not distinguish between short-acting or long-acting therapy. Lastly, DCE exercises may differ from real-world situations because the selected attributes may simplify true real-world scenarios; study respondents had equal information about alternative treatment options; there were no supply or budget constraints in our model, and not all options may be available to all individuals with HAE in clinical practice.

In conclusion, effectiveness is the primary factor in LTP treatment selection by individuals with HAE, but it is not the only aspect patients consider when offered LTP to manage their HAE. Individuals with HAE are open to changing their LTP, especially if they are experiencing challenges with disease or treatment burden. Because individuals with HAE also consider convenience and the route of administration important, HCPs should consider these attributes when tailoring individualized treatment recommendations to their patients. Assessing the preferences of individual patients is a critical element of shared decision-making in HAE LTP treatment decisions.

Data Availability

All data supporting the conclusions of these analyses are presented in the manuscript or the supporting information. Additional details can be obtained from the corresponding author on reasonable request.

Acknowledgments

The authors thank Rebecca Hahn, MPH, CMPP, of KJT Group, Inc. (Rochester, NY) for providing medical writing support, funded by BioCryst Pharmaceuticals, Inc. (Durham, NC) in accordance with Good Publication Practice (GPP 2022) and International Committee of Medical Journal Editors (ICMJE) guidelines. The authors also thank Diane Ramsey-Paige, BSN, PNP, CCRC, of BioCryst Pharmaceuticals, Inc. (Durham, NC), for her input on the graphical abstract.

Disclosures

Dr Soteres is a consultant for BioCryst Pharmaceuticals, Inc. Dr Olson is a consultant for BioCryst Pharmaceuticals, Inc. Dr Poulos is an employee of RTI Health Solutions, an independent nonprofit research organization retained by BioCryst to provide consultation for this research. Drs Gillard and Nestler-Parr and Ms Ray are employees of and may own stock/stock options in BioCryst Pharmaceuticals, Inc. When the research was conducted and the manuscript was developed, Dr Lopez-Gonzalez was an employee of BioCryst Pharmaceuticals, Inc; she is currently employed by Syndax Pharmaceuticals. Mr Lionetti is employed by KJT Group, Inc, a consulting company that received funding from BioCryst Pharmaceuticals, Inc, to conduct this research.

Funding

This work was supported by BioCryst Pharmaceuticals, Inc.

Supplementary Data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.anai.2025.12.011.

References

1. Johnston DT. Diagnosis and management of hereditary angioedema. *J Am Osteopath Assoc*. 2011;111(1):28–36.
2. Longhurst HJ, Bork K. Hereditary angioedema: an update on causes, manifestations and treatment. *Br J Hosp Med (Lond)*. 2019;80(7):391–398.
3. Zafra H. Hereditary angioedema: a review. *WMJ*. 2022;121(1):48–53.
4. Guan X, Sheng Y, Liu S, He M, Chen T, Zhi Y. Epidemiology, economic, and humanistic burden of hereditary angioedema: a systematic review. *Orphanet J Rare Dis*. 2024;19(1):256.
5. Tutunaru CV, Ică OM, Mitroi GG, Neagoe CD, Mitroi GF, Orzan OA, et al. Unveiling the complexities of hereditary angioedema. *Biomolecules*. 2024;14(10):1298.
6. Aygören-Pürsün E, Bygum A, Beusterien K, Hautamaki E, Sisic Z, Boysen HB, et al. Estimation of EuroQol 5-Dimensions health status utility values in hereditary angioedema. *Patient Prefer Adherence*. 2016;10:1699–1707.
7. Lumry WR, Settipane RA. Hereditary angioedema: epidemiology and burden of disease. *Allergy Asthma Proc*. 2020;41(Suppl 1):S08–S13.
8. Bygum A, Aygören-Pürsün E, Beusterien K, Hautamaki E, Sisic Z, Wait S, et al. Burden of illness in hereditary angioedema: a conceptual model. *Acta Derm Venereol*. 2015;95(6):706–710.
9. Lumry WR, Castaldo AJ, Vernon MK, Blaustein MB, Wilson DA, Horn PT. The humanistic burden of hereditary angioedema: impact on health-related quality of life, productivity, and depression. *Allergy Asthma Proc*. 2010;31(5):407–414.
10. Caballero T, Aygören-Pürsün E, Bygum A, Beusterien K, Hautamaki E, Sisic Z, et al. The humanistic burden of hereditary angioedema: results from the Burden of Illness Study in Europe. *Allergy Asthma Proc*. 2014;35(1):47–53.
11. Fouche AS, Saunders EF, Craig T. Depression and anxiety in patients with hereditary angioedema. *Ann Allergy Asthma Immunol*. 2014;112(4):371–375.
12. Aygören-Pürsün E, Bygum A, Beusterien K, Hautamaki E, Sisic Z, Wait S, et al. Socioeconomic burden of hereditary angioedema: results from the hereditary angioedema burden of illness study in Europe. *Orphanet J Rare Dis*. 2014;9:99.
13. Cardarelli WJ. Economic burden limiting proper healthcare delivery, management, and improvement of patient outcomes. *Am J Manag Care*. 2018;24(14):S308–S313. (suppl).
14. Riedl MA, Banerji A, Gower R. Current medical management of hereditary angioedema: follow-up survey of US physicians. *Ann Allergy Asthma Immunol*. 2021;126(3):264–272.
15. Lamacchia D, Nappi E, Marzio V, Locatelli F, Messina MR, Heffler E. Hereditary angioedema: current therapeutic management and future approaches. *Curr Opin Allergy Clin Immunol*. 2024;24(4):257–265.
16. Costanzo G, Sambugaro G, Firinu D. Hereditary angioedema due to C1-inhibitor deficiency: current therapeutic approaches. *Curr Opin Allergy Clin Immunol*. 2024;24(6):488–495.
17. Betschel S, Badiou J, Binkley K, Borici-Mazi R, Hébert J, Kanani A, et al. The international/Canadian hereditary angioedema guideline. *Allergy Asthma Clin Immunol*. 2019;15:72.
18. Busse PJ, Christiansen SC, Riedl MA, Banerji A, Bernstein JA, Castaldo AJ, et al. US HAEA Medical Advisory Board 2020 guidelines for the management of hereditary angioedema. *J Allergy Clin Immunol Pract*. 2021;9(1):132–150. e3.
19. Maurer M, Magerl M, Betschel S, Aberer W, Ansotegui IJ, Aygören-Pürsün E, et al. The international WAO/EAACI guideline for the management of hereditary angioedema—the 2021 revision and update. *Allergy*. 2022;77(7):1961–1990.
20. Riedl MA, Banerji A, Busse PJ, Johnston DT, Davis-Lorton MA, Patel S, et al. Patient satisfaction and experience with intravenously administered C1-inhibitor concentrates in the United States. *Ann Allergy Asthma Immunol*. 2017;119(1):59–64.
21. Riedl MA, Craig TJ, Banerji A, Aggarwal K, Best JM, Rosselli J, et al. Physician and patient perspectives on the management of hereditary angioedema: a survey on treatment burden and needs. *Allergy Asthma Proc*. 2021;42(3):S17–S25.
22. Radojicic C, Riedl MA, Craig TJ, Best JM, Rosselli J, Hahn R, et al. Patient perspectives on the treatment burden of injectable medication for hereditary angioedema. *Allergy Asthma Proc*. 2021;42(3):S4–S10.
23. Do T, Riedl MA. Current and emerging therapeutics in hereditary angioedema. *Immunol Allergy Clin North Am*. 2024;44(3):561–576.
24. Zuraw BL, Banerji A, Bernstein JA, Busse PJ, Christiansen SC, Davis-Lorton M, et al. US Hereditary Angioedema Association Medical Advisory Board 2013 recommendations for the management of hereditary angioedema due to C1 inhibitor deficiency. *J Allergy Clin Immunol Pract*. 2013;1(5):458–467.
25. Zuraw BL, Bernstein JA, Lang DM, Craig T, Dreyfus D, Hsieh F, et al. A focused parameter update: hereditary angioedema, acquired C1 inhibitor deficiency, and angiotensin-converting enzyme inhibitor-associated angioedema. *J Allergy Clin Immunol*. 2013;131(6):1491–1493.
26. Craig T, Busse P, Gower RG, Johnston DT, Kashkin JM, Li HH, et al. Long-term prophylaxis therapy in patients with hereditary angioedema with C1 inhibitor deficiency. *Ann Allergy Asthma Immunol*. 2018;121(6):673–679.
27. Dorman S, McNeil D, Nelson J, Howard L, Ambler W, Bailey T, et al. Understanding the impact of long-term prophylaxis switches for patients with hereditary angioedema. *J Allergy Clin Immunol*. 2024;153(2). AB362.
28. Riedl MA, Soteres D, Sublett JW, Desai B, Tomita D, Collis P, et al. Hereditary angioedema outcomes in US patients switched from injectable long-term prophylactic medication to oral berotralstat. *Ann Allergy Asthma Immunol*. 2024;132(4):505–511. e1.
29. Magerl M, Martinez-Saguer I, Schauf L, Pohl S, Brendel K. The current situation of hereditary angioedema patients in Germany: results of an online survey. *Front Med (Lausanne)*. 2024;10:1274397.
30. Bork K, Anderson JT, Caballero T, Craig T, Johnston DT, Li HH, et al. Assessment and management of disease burden and quality of life in patients with hereditary angioedema: a consensus report. *Allergy Asthma Clin Immunol*. 2021;17(1):40.
31. Greve J, Kinaciyan T, Maurer M, Dillenburger B, Recke A, Schöffl C. Expert consensus on prophylactic treatment of hereditary angioedema. *Allergo J Int*. 2022;31(7):233–242.
32. Riedl MA, Neville D, Cloud B, Desai B, Bernstein JA. Shared decision-making in the management of hereditary angioedema: an analysis of patient and physician perspectives. *Allergy Asthma Proc*. 2022;43(5):397–405.
33. Settipane R, Bukstein D, Riedl MA. Hereditary angioedema and shared decision making. *Allergy Asthma Proc*. 2020;41(suppl 1):S55–S60.
34. Odin R, Anderson J, Jacobs J, Jones D, Li HH, Lumry W, et al. The utility of shared decision making in the management of hereditary angioedema. *Allergy Asthma Proc*. 2024;45(6):434–437.
35. Barbosa CD, Balp MM, Kulich K, Germain N, Rofail D. A literature review to explore the link between treatment satisfaction and adherence, compliance, and persistence. *Patient Prefer Adherence*. 2012;6:39–48.
36. Swiatoniowska-Lonc N, Tanski W, Polanski J. Jankowska-Polanska B. Treatment satisfaction and pharmacological adherence in patients with chronic diseases. *Eur J Cardiovasc Nurs*. 2022;21(suppl 1). zvac060.067.
37. Geba D, Mohd Sani J, Gascon M, Hahn R, Aggarwal K, Rosselli J. Hereditary angioedema patients would prefer newer-generation oral prophylaxis. *J Drug Assess*. 2021;10(1):51–56.
38. Anderson J, Levy DS, Lumry W, Koochaki P, Lanar S, Henry Li H. Letting the patients speak: an in-depth, qualitative research-based investigation of factors relevant to health-related quality of life in real-world patients with hereditary angioedema using subcutaneous C1 inhibitor replacement therapy. *Allergy Asthma Clin Immunol*. 2021;17(1):60.
39. Bridges JF, Hauber AB, Marshall D, Lloyd A, Prosser LA, Regier DA, et al. Conjoint analysis applications in health—a checklist: a report of the ISPOR Good Research Practices for Conjoint Analysis Task Force. *Value Health*. 2011;14(4):403–413.
40. Campoamor NB, Guerrini CJ, Brooks WB, Bridges JFP, Crossnohere NL. Pretesting discrete-choice experiments: a guide for researchers. *Patient*. 2024;17(2):109–120.
41. Soekhai V, Whichello C, Levitan B, Veldwijk J, Pinto CA, Donkers B, et al. Methods for exploring and eliciting patient preferences in the medical product lifecycle: a literature review. *Drug Discov Today*. 2019;24(7):1324–1331.
42. US Food and Drug Administration. Incorporating voluntary patient preference information over the total product life cycle: draft guidance for industry, Food and Drug Administration staff, and other interested parties. Accessed November 15, 2024. Available at: https://www.fda.gov/media/181509/download.
43. The PREFER Consortium. PREFER Recommendations: why, when and how to assess and use patient preferences in medical product decision-making. Accessed December 17, 2024. Available at: https://zenodo.org/records/6592304.
44. Yang JC, Johnson FR, Kilambi V, Mohamed AF. Sample size and utility-difference precision in discrete-choice experiments: a meta-simulation approach. *J Choice Modell*. 2015;16:50–57.
45. de Bekker-Grob EW, Donkers B, Jonker MF, Stolk EA. Sample size requirements for discrete-choice experiments in healthcare: a practical guide. *Patient*. 2015;8(5):373–384.
46. World Health Organization. The World Health Organization-Five Well-Being Index (WHO-5). Accessed November 15, 2024. Available at: https://www.who.int/publications/m/item/WHO-UCN-MSD-MHE-2024.01.
47. Vanya M, Watt M, Shahraz S, Kosmas CE, Rhoten S, Costa-Cabral S, et al. Content validation and psychometric evaluation of the Angioedema Quality of Life Questionnaire for hereditary angioedema. *J Patient Rep Outcomes*. 2023;7(1):33.
48. Hollin IL, Paskett J, Schuster ALR, Crossnohere NL, Bridges JFP. Best–worst scaling and the prioritization of objects in health: a systematic review. *Pharmacoeconomics*. 2022;40(9):883–899.
49. Schuster ALR, Crossnohere NL, Campoamor NB, Hollin IL, Bridges JFP. The rise of best-worst scaling for prioritization: a transdisciplinary literature review. *J Choice Modell*. 2024;50:100466.
50. Mühlbacher AC, de Bekker-Grob EW, Rivero-Arias O, Levitan B, Vass C. How to present a decision object in health preference research: attributes and levels, the decision model, and the descriptive framework. *Patient*. 2025;18(4):291–302.
51. Noben CYG, Stammen LA, Vaassen S, Haeren R, Stassen L, van Mook W, et al. Discrete choice experiment on educating value-based healthcare. *Postgrad Med J*. 2021;97(1150):515–520.
52. Prochaska JO, Velicer WF. The transtheoretical model of health behavior change. *Am J Health Promot*. 1997;12(1):38–48.
53. Ajzen I. From intentions to actions: a theory of planned behavior. In: Kuhl J, Beckmann J, eds. *Action Control: From Cognition to Behavior*. Springer; 1985:11–39.
54. Fishbein M. A reasoned action approach to health promotion. *Med Decis Making*. 2008;28(6):834–844.
55. Montori VM, Ruissen MM, Hargraves IG, Brito JP, Kunneman M. Shared decision-making as a method of care. *BMJ Evid Based Med*. 2023;28(4):213–217.
56. National Institute for Health and Care Excellence (NICE). Shared decision making. Accessed November 15, 2024. Available at: https://www.nice.org.uk/guidance/ng197.
57. US Food and Drug Administration. Patient-focused drug development: collecting comprehensive and representative input guidance for industry, Food and Drug

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Administration staff, and other stakeholders. Accessed November 15, 2024. Available at: https://www.fda.gov/media/139088/download.
58. Gower RG. HAE update: determining optimal patient specific therapy. *Allergy Asthma Proc*. 2013;34(1):7–12.
59. Jose J, Lehman EB, Craig T. Evaluating satisfaction of patients with hereditary angioedema with their past and present treatments: implications for future therapies. *Allergy Asthma Proc*. 2018;39(1):74–80.
60. Center for Biologics Evaluation and Research, US Food and Drug Administration. The voice of the patient: hereditary angioedema – report of the U.S. Food and Drug Administration patient-focused drug development initiative public meeting September 2017. Accessed November 15, 2024. Available at: https://www.fda.gov/files/about%20fda/published/The-Voice-of-the-Patient—Hereditary-Angioedema.pdf.
61. Anderson J, Maina N. Reviewing clinical considerations and guideline recommendations of C1 inhibitor prophylaxis for hereditary angioedema. *Clin Transl Allergy*. 2022;12(1):e12092.
62. Smith TD, Riedl MA. The future of therapeutic options for hereditary angioedema. *Ann Allergy Asthma Immunol*. 2024;133(4):380–390.
63. Jain G, Walter L, Reed C, O'Donnell P, Troy J. How do patients and physicians communicate about hereditary angioedema in the United States? *PLoS One*. 2021;16(12):e0260805.
64. Banerji A, Anderson J, Johnston DT. Optimal management of hereditary angioedema: shared decision-making. *J Asthma Allergy*. 2021;14:119–125.
65. Paige D, Maina N, Anderson JT. Hereditary angioedema: comprehensive management plans and patient support. *Allergy Asthma Proc*. 2020;41(suppl 1):S38–S42.
66. The Earnest Analytics Company. Orladeyo: a tough pill to swallow for competitors in the HAE space. Accessed December 17, 2024. Available at: https://www.earnestanalytics.com/insights/all-posts/orladeyo-a-tough-pill-to-swallow-for-competitors-in-the-hae-space#:~:text=At%20a%20recent%20conference%2C%20BCRX,85%25%20of%20HAE%20prescription%20volume.

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eTable 1
Correlation Between Agreement With HAE Impact Statements and Willingness to Switch Long-Term Prophylaxis (N = 150)

Impact statement	Pearson correlation coefficient
HAE defines my life.	0.0173
HAE is a lower priority compared to other conditions that I have.	−0.0643
Having HAE takes a serious toll on my mental health.	0.0647
I have a very good quality of life, despite my HAE attacks.	−0.0444
I have sought urgent/emergency care often in the past 2 years because of HAE attacks.	0.0468
I often worry about my HAE.	0.1832[a]
HAE affects my relationships with others.	0.0714
HAE makes me feel isolated and lonely.	0.0069
HAE significantly affects my ability to succeed at work/school/other endeavors.	0.0354
I am concerned with my physical appearance because of my HAE.	0.0636
I am still socially active despite my HAE.	0.0700
My hobbies have been impacted due to my HAE.	0.0445
I have missed a significant amount of time at work/school in the past 2 years because of my HAE.	0.0215
My ability to travel is impacted by my HAE.	0.1355

Abbreviation: HAE, hereditary angioedema.
[a]*P* is less than .05.

eTable 2
Impact of Behavioral Intention Analysis Domain Items on Willingness to Switch Long-Term Prophylaxis (N = 150)

Group	Variable	Metric	Value (%)	Statistical significance	Coefficient
Perceived control model	My insurance is likely to make it challenging to switch to a new HAE medication.	Relative impact	1	—	−0.3202
	I am scared that my HAE may not be as well controlled on another HAE medication.	Relative impact	3	—	0.5250
	Control variable: attitudinal behavior composite score	Relative impact	84	[d]	−1.5672
	Control variable: social norms composite score	Relative impact	12	[c]	0.3856
Behavioral attitudes model	Oral HAE medications are just as effective as injectable HAE medications.[a]	Relative impact	23	[c]	−1.6750
	Oral HAE medications are more convenient than injectable HAE medications.[a]	Relative impact	16	[b]	−1.6376
	I feel confident that I can take a daily HAE medication.[a]	Relative impact	2	—	−5.43E-01
	I do not believe I could achieve greater efficacy with a different, currently available HAE medication.	Relative impact	6	—	−0.8712
	It took a long time to get my HAE under control; it would be risky to switch my medication now.	Relative impact	2	—	0.4305
	I do not believe I could achieve a better quality of life with a different, currently available HAE medication.	Relative impact	0	—	1.02E-02
	My quality of life is acceptable enough to prevent me from taking the risk to switch to a different medication.	Relative impact	30	[c]	−1.6781
	The hassle of switching my HAE medication is not worth a potential improvement in my quality of life.	Relative impact	4	—	−0.6494
	Control variable: perceived control composite score	Relative impact	4	—	−0.1993
	Control variable: social norms composite score	Relative impact	13	[b]	0.3163
Social norms model	I like to take the same HAE medication as my family.	Relative impact	2	—	0.6994
	If my family had success with a medication, I would be interested in switching to it.	Relative impact	1	—	0.3813
	If another patient with HAE shared a positive experience on a medication (eg, support group, Facebook group), I would be interested in switching to it.	Relative impact	9	[b]	1.4961
	Control variable: attitudinal behavior composite score	Relative impact	0	—	−0.0092
	Control variable: perceived control composite score	Relative impact	87	[c]	−1.5179

Abbreviation: HAE, hereditary angioedema.
[a]Item was reverse-coded.
[b]*P* is less than .05.
[c]*P* is less than .01.
[d]*P* is less than .001.